

Problem

Design a problem to help other students better understand finding the total energy in a given signal.

Step-by-step solution

Step 1 of 2

Consider a signal $f(t) = Ae^{-kt}u(t)$

Where A, k are constants

The total energy in the given signal is

$$W_{\Omega} = \int_{-\infty}^{\infty} f^2(t) dt$$

$$W_{\Omega} = \int_{-\infty}^{\infty} [Ae^{-kt}u(t)]^2 dt$$

$$W_{\Omega} = \int_0^{\infty} [Ae^{-kt}]^2 dt$$

Step 2 of 2

$$W_{\Omega} = A^2 \int_0^{\infty} e^{-2kt} dt$$

$$W_{\Omega} = A^2 \left[\frac{e^{-2kt}}{-2k} \right]_0^{\infty}$$

$$W_{\Omega} = A^2 \left[\frac{e^{-\infty} - e^0}{-2k} \right]$$

$$W_{\Omega} = \frac{A^2}{-2k} [0 - 1]$$

$$\therefore \boxed{W_{\Omega} = \frac{A^2}{2k} \text{ J}}$$